AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

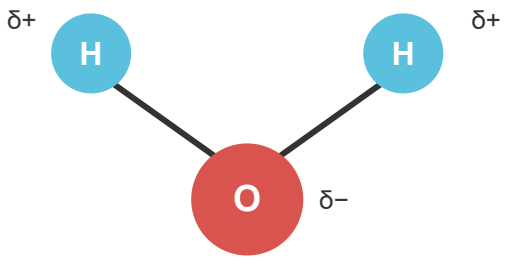
Chemistry for biologists

Living things are built from a small set of molecules, and most of their behaviour follows from the shapes of those molecules and the forces between them. This topic covers water, carbohydrates, lipids and proteins, and explains how the structure of each one fits its job in an organism.

1. Water: a polar molecule

A water molecule is made of one oxygen atom covalently bonded to two hydrogen atoms. The shared electrons are not held equally. Oxygen is much more electronegative than hydrogen, so it pulls the bonding electrons towards itself. The oxygen end becomes slightly negative (δ−) and the two hydrogen ends become slightly positive (δ+).

The molecule is bent rather than straight, at roughly 104.5 degrees. Because of this shape the small charges do not cancel out, so the whole molecule behaves as a dipole. This is the single most important fact about water, because nearly every property discussed below depends on it.



δ− on oxygen, δ+ on each hydrogen

Why water is polar

1. Oxygen has a higher electronegativity than hydrogen.
2. The shared bonding electrons spend more time near the oxygen nucleus.

   Electrons are pulled towards the atom that attracts them more strongly.
3. Oxygen becomes δ− and each hydrogen becomes δ+.
4. The molecule is bent, so the two bond dipoles add up instead of cancelling.

   A straight molecule with the same bonds would have no overall dipole.
5. The result is a permanent dipole, which lets water form hydrogen bonds with other polar molecules.

- **Covalent bond:** a shared pair of electrons between two atoms, the strong bond holding the O and H atoms of one water molecule together.
- **Hydrogen bond:** a weak attraction between a δ+ hydrogen on one molecule and a δ− oxygen or nitrogen on another. It is much weaker than a covalent bond but there are very many of them.
- **Polar molecule:** a molecule with an uneven charge distribution, so it has a positive end and a negative end.

2. Why water is essential for life

Water is the medium in which almost all biochemistry happens, and it is the most abundant compound in most cells. Its properties are not accidents. Hydrogen bonding between water molecules holds them together, and this gives water a high boiling point and a high specific heat capacity, and makes it act as a solvent, a transport medium and a thermal buffer.

Water as a solvent

Ionic compounds such as sodium chloride dissolve because water molecules surround the separated ions. The δ− oxygens cluster around Na+ and the δ+ hydrogens cluster around Cl−, each ion being wrapped in a shell of water and kept apart. Polar covalent molecules such as glucose dissolve in the same way, forming hydrogen bonds with water. Non-polar substances such as fats do not, because they cannot compete with the hydrogen bonds between water molecules. They are hydrophobic and separate out from the water.

This is why water is a good transport medium. Sugars, amino acids, ions and gases can be carried in the blood, the cytoplasm and the xylem and phloem sap because they are dissolved in water. Dissolved substances also stay available for reactions, because they can move and collide freely.

Cohesion, adhesion and surface tension

Water molecules are attracted to each other by hydrogen bonds, which is called cohesion. Molecules in the middle of a liquid are pulled equally in all directions. Molecules at the surface are pulled only sideways and downwards, so the surface is pulled tight. This is surface tension, which lets small insects such as pond skaters walk on the surface without breaking it.

Water is also attracted to other polar surfaces such as the walls of xylem vessels. This is adhesion. Cohesion keeps a continuous column of water together under tension during transpiration, and adhesion stops the column from pulling away from the vessel walls.

Specific heat capacity and latent heat

Specific heat capacity is the energy needed to raise the temperature of 1 kg of a substance by 1 °C. Water has a high value, about 4200 J kg^{-1} $°C^{-1}$. A large part of the energy input goes into breaking hydrogen bonds before the molecules can move faster, so the temperature rises slowly. This stabilises the temperature of lakes, seas and the cytoplasm of cells.

Latent heat of vaporisation is the energy needed to change 1 kg of a liquid into a gas at a constant temperature. Water has a high value because a lot of energy is needed to break the hydrogen bonds holding the liquid together.

When water evaporates from the skin, the most energetic molecules escape and the remaining water is cooler. This is evaporative cooling, the basis of sweating.

Property	Why it happens	Biological significance
High specific heat capacity	Energy is absorbed breaking hydrogen bonds	Stable temperature in organisms and aquatic habitats
High latent heat of vaporisation	Many hydrogen bonds must be broken to evaporate	Cooling by sweating and transpiration
High surface tension	Cohesion pulls surface molecules inwards	Insects walk on water; capillary action in plants
Density maximum at 4 °C; ice less dense than liquid	Hydrogen bonds hold molecules apart in ice	Ice floats, insulating the water below so aquatic life survives winter
Universal solvent	Polar and charged molecules are attracted to water	Transport and a medium for reactions

KEY IDEA

Ice is less dense than liquid water because hydrogen bonds hold the molecules in an open lattice. Most substances become denser when they freeze, but water does not, which is why ponds freeze from the top down.

COMMON MISTAKE

Do not confuse specific heat capacity with latent heat. Specific heat is about a temperature change within one state. Latent heat is about a change of state at constant temperature.

3. Carbohydrates

Carbohydrates contain carbon, hydrogen and oxygen, with hydrogen and oxygen in a 2:1 ratio. Their general formula is $C_n(H_2O)_n$. They are the main energy source for cell respiration and, as polymers, also serve as stores of energy. They are classified by the number of sugar units they contain: monosaccharides, disaccharides and polysaccharides.

Monosaccharides

Monosaccharides are the simplest sugars, with three to seven carbon atoms. The ones you need to know are glucose, fructose and galactose, all of which have the formula $C_6H_{12}O_6$ but different structures. Glucose exists in two ring forms, alpha and beta, which differ only in the position of the –OH group on carbon 1. Only alpha-glucose is needed for this topic, and it is the form that makes starch and glycogen. Monosaccharides are soluble, sweet and reducing sugars, and they are small enough to pass through cell membranes by facilitated diffusion or active transport.

EXAM TIP

When a question asks why glucose is soluble, give the reason in terms of the structure: the many –OH groups make glucose polar, so it forms hydrogen bonds with water.

Disaccharides and the glycosidic bond

Two monosaccharides join by a condensation reaction. An –OH group from one sugar combines with an –H from the other, and a molecule of water is released. The new covalent link is a glycosidic bond. Hydrolysis is the reverse: a water molecule is added to break the bond. Enzymes such as maltase, sucrase and lactase carry out these hydrolysis reactions during digestion.

- **Maltose:** two alpha-glucose units joined by a 1,4-glycosidic bond. It is formed when starch is broken down, for example during germination.
- **Sucrose:** alpha-glucose joined to fructose by a 1,2-glycosidic bond. It is the transport sugar in plants and the sugar most people eat.
- **Lactose:** galactose joined to glucose. It is the main sugar in milk and is hydrolysed by lactase.

Build maltose, and then a longer chain, one condensation reaction at a time.

Polysaccharides

Polysaccharides are long chains of many monosaccharide units joined by glycosidic bonds, typically several hundred to thousands of units long. Their size makes them insoluble, so they do not affect the water potential of the cell and cannot diffuse out of it. This is why they are good storage molecules: a cell can store a large amount of glucose as starch or glycogen without a huge osmotic effect.

Starch and glycogen

Starch is the storage polysaccharide of plants. It is a mixture of two polymers of alpha-glucose. Amylose is an unbranched chain of 1,4-linked units that coils into a helix, which makes it compact. Amylopectin is branched, with 1,6-glycosidic bonds appearing about every 20 to 30 glucose units, and the branches end in many terminal glucose units.

Glycogen is the storage polysaccharide of animals and fungi. Its structure is like amylopectin but with more frequent branches, roughly every 8 to 10 units. Many branches mean many free ends, so enzymes can add or remove glucose from many points at once. Glycogen can therefore be broken down quickly, which suits animal muscle and liver during intense activity.

Why glycogen suits animal energy storage

1. Glycogen is a polymer of alpha-glucose, so it stores a lot of energy in a small volume.

2. It is insoluble and does not create an osmotic effect, so the cell does not take in extra water.

   Dissolved glucose molecules would each add to the solute concentration.

3. Its many branches give many non-reducing ends.

4. Enzymes can release glucose from every end at once, so glucose is released rapidly when needed.

   Rapid glucose release supports the high demand for respiration during exercise.

Feature	Monosaccharide	Disaccharide	Polysaccharide
Number of units	1	2	Many (often hundreds or more)
Glycosidic bonds	None	1	Many
Solubility	Soluble	Soluble	Insoluble
Sweet taste	Yes	Yes	No
Osmotic effect	Yes	Yes	Little or none
Examples	Glucose, fructose, galactose	Maltose, sucrose, lactose	Starch, glycogen

4. Lipids

Lipids are a varied group of molecules that are insoluble in water and soluble in organic solvents. They are not polymers, because they are not made of many identical repeating units. They contain much less oxygen than carbohydrates, so when they are oxidised they release much more energy per gram. The main groups are triglycerides, phospholipids and steroids such as cholesterol.

Triglycerides

A triglyceride is made from one glycerol molecule and three fatty acids. Each fatty acid has a carboxyl group at one end and a long hydrocarbon tail. The carboxyl group of each fatty acid reacts with an –OH group of glycerol in a condensation reaction, forming an ester bond. Three water molecules are released. Hydrolysis of an ester bond reverses this, which is how lipase digests fat.

Saturated and unsaturated fatty acids

In a saturated fatty acid all the carbon–carbon bonds in the tail are single bonds, so the tail is straight. Straight tails pack closely together, with many van der Waals forces between them, so fats made from them are solid at room temperature. This is why butter is solid.

An unsaturated fatty acid has at least one carbon–carbon double bond. A double bond introduces a kink, so the tails cannot pack tightly. Fewer intermolecular forces act between them, so less energy is needed to separate them, and oils are liquid at room temperature. A polyunsaturated fat has more than one double bond.

COMMON MISTAKE

A double bond between carbons does not mean extra hydrogens are present. Unsaturated means there is less hydrogen than the maximum possible. The important point for melting point is the kink, not the hydrogen count.

Phospholipids and functions of lipids

A phospholipid has two fatty acid tails and a phosphate group in place of the third fatty acid. The phosphate head is polar and hydrophilic, while the tails are non-polar and hydrophobic. This dual character is what makes phospholipids form the bilayer of cell membranes, which is covered in more detail in the cell membrane topic.

The main functions of lipids are:

- **Energy storage:** a gram of fat releases about 9 kcal when oxidised, compared with about 4 kcal from a gram of carbohydrate or protein, because fat has far more C–H bonds and less oxygen.
- **Insulation:** fat under the skin reduces heat loss, and whale blubber is a good example.
- **Protection:** fat around vital organs cushions them from physical damage.
- **Buoyancy:** lipids are less dense than water, which helps aquatic animals float.
- **Waterproofing:** waxes and cuticles on leaves and skin stop water loss.

Why saturated fats are solid at room temperature

1. A saturated fatty acid tail has only single bonds, so it is straight.
2. Straight tails lie close together and their van der Waals forces add up across the whole length.
3. A lot of energy is needed to pull the tails apart.

   More energy is needed to overcome the forces, so the melting point is higher.
4. At room temperature the fat stays solid.

5. Proteins

Proteins are polymers of amino acids and are found in every cell. They do many jobs: enzymes catalyse reactions, transport proteins move substances, structural proteins give support, and antibodies defend the body. What a protein does depends on its three-dimensional shape, and that shape comes from the order of its amino acids.

Amino acids and the peptide bond

Every amino acid has the same core: a central carbon attached to an amine group (–NH_2), a carboxyl group (–COOH), a hydrogen and a variable R group. The R group differs between the twenty common amino acids and determines the properties of each one.

Amino acids join by condensation. The –COOH group of one amino acid reacts with the –NH_2 group of the next, forming a peptide bond (–CO–NH–) and releasing water. A chain of many amino acids joined this way is a polypeptide. Hydrolysis of peptide bonds breaks the chain again.

Four levels of protein structure

- **Primary structure:** the sequence of amino acids in the chain. This is determined by the gene coding for the protein.
- **Secondary structure:** local folding of the chain into an alpha helix or a beta pleated sheet, held in place by hydrogen bonds between the C=O of one residue and the N–H of another. These bonds are along the backbone, not the R groups.
- **Tertiary structure:** the overall three-dimensional folding, stabilised by interactions between R groups. These include ionic bonds, hydrogen bonds, disulphide bridges between cysteine residues and hydrophobic interactions.
- **Quaternary structure:** two or more polypeptide chains held together, as in haemoglobin.

Why changing one amino acid can change a protein's function

1. Changing one amino acid changes the R group at that position.
2. The new R group may have different charge, size or polarity.

   Only the R group varies, so the change is in its chemistry.
3. This alters the bonds that form between R groups, so the tertiary structure changes.
4. The shape of the active or binding site changes, so the protein may no longer bind its substrate.

Fibrous and globular proteins

Globular proteins are folded into compact, roughly spherical shapes. Their hydrophilic R groups point outwards, so they dissolve in water, and they can carry substances or act as enzymes. Haemoglobin is a globular protein with four subunits, each containing a haem group. Fibrous proteins have long chains arranged in parallel and held together by many cross-links. They are insoluble and strong, which suits structural roles. Collagen is a fibrous protein with three polypeptide chains wound into a triple helix. Each strand is coiled tightly, and this gives collagen a high tensile strength for tendons and skin.

Feature	Globular protein	Fibrous protein
Overall shape	Compact, roughly spherical	Long and thread-like
Solubility in water	Soluble	Insoluble
Main role	Transport, enzymes, signalling	Structural support
Example	Haemoglobin, insulin	Collagen, keratin

Denaturation

Heat, extreme pH, high salt concentration and some organic solvents break the hydrogen bonds, ionic bonds and disulphide bridges that maintain the tertiary and secondary structure. The chain unfolds and the shape of the active site is lost. The primary structure is not broken, because peptide bonds are unaffected, so the protein is denatured but the sequence stays the same. Denaturation is usually irreversible for enzymes.

KEY IDEA

Protein function depends on shape, shape depends on the sequence and on the bonds between R groups, and denaturation destroys the shape without breaking the peptide bonds.

EXAM TIP

When you explain a protein's shape, name the bond type and the level of structure. Saying "the bonds" is not enough for full marks.

Quick checks

QUICK CHECK

Which property of water is mainly responsible for the cooling effect of sweating?

- High specific heat capacity
- High latent heat of vaporisation
- High density at 4 °C
- Universal solvent

Sweat evaporates, and evaporation needs a large amount of energy to break hydrogen bonds. The energy comes from the skin, so the skin cools. Specific heat capacity describes temperature change inside a liquid, so it is the commonest wrong answer.

QUICK CHECK

A disaccharide is formed from two alpha-glucose molecules. What is released when the glycosidic bond forms?

- Carbon dioxide
- Oxygen
- Water
- A hydrogen molecule

Condensation joins two monosaccharides by combining an –OH group with an –H, and the two form water. Hydrolysis is the reverse. Students often think a bond forms by losing hydrogen gas, which is incorrect.

QUICK CHECK

Why is glycogen a better energy store in animal muscle than a single glucose molecule?

- It has a higher solubility in water
- It releases glucose more slowly
- It is insoluble and has many ends for rapid glucose release
- It contains more oxygen than glucose

Glycogen is insoluble, so it does not create a large osmotic effect, and its branches give many terminal glucose units for enzymes to act on at once. A common wrong answer is that glycogen is more soluble; it is actually the opposite.

QUICK CHECK

Which change would most likely lower the melting point of a fat?

- Making the fatty acid tails longer and fully saturated
- Introducing carbon–carbon double bonds into the fatty acid tails
- Joining the fatty acids to a larger glycerol molecule
- Increasing the number of hydrogen atoms in the tails

Double bonds create kinks that stop the tails packing together, which weakens the van der Waals forces and lowers the melting point. Longer saturated tails would pack more tightly and raise the melting point. Adding hydrogen to a tail is saturation, which raises the melting point.